

Project MediLong

Backend & Cloud Squad — Detailed Report

Squad Report · 27-Day Sprint · Sept 2 – Sept 28

Squad	Backend & Cloud
Members	Sridhar, Murasoli, Kamalesh G
Sprint Window	Sept 2 – Sept 28, 2026 (27-day sprint)

Mission

Own the FastAPI service, Firebase schema, secure upload buckets, and orchestration that routes data between all three pillars and the decision core. Every other squad's output passes through here.

Contract Slice Owned

Owns the unified JSON contract document itself and enforces it at the API boundary — request/response validation for every route.

Sprint-by-Sprint Tasks

Sprint 1 (Sept 2–8)

- Publish and lock the unified JSON contract by Sept 3; circulate to all squad leads.
- FastAPI skeleton: /intake, /upload/ultrasound, /upload/lab-pdf, /predict — all returning mocked, contract-shaped responses.
- Firebase schema design (patients, uploads, predictions collections) and secure bucket config for image/PDF storage.

Sprint 2 (Sept 9–15)

- Replace mocks with real Firebase persistence.
- Auth/session handling for uploads (a lightweight token scheme is fine for a prototype).
- Define internal service contracts to the Data Pipelines and ML Fusion squads (function signatures / internal REST calls).

Sprint 3 (Sept 16–22)

- Wire /predict to call all three pipelines live, fuse outputs into the contract shape, and forward to the decision core.
- Structured, signed PDF summary generation service.
- Daily integration checkpoints with all squads — highest-risk week for silent contract drift.

Sprint 4 (Sept 23–28)

- Async I/O audit for latency (< 3s end-to-end target) — profile OCR and CV calls first, they're the likely bottlenecks.
- Error handling for corrupted uploads, malformed PDFs, timeout fallbacks.

Dependencies

- Blocked by: nothing structurally — Backend can mock everyone in Sprint 1.
- Blocks: Frontend (live endpoints), all pipeline squads (needs their real output format confirmed by Sprint 2).

Definition of Done

- All routes contract-compliant, no schema drift from Day 2 lock.
- End-to-end latency under 3 seconds.
- PDF export signed and reproducible.